

COMP-3150 CHAPTERS 1 TO 3: LECTURE NOTES

(Chs 2 and 3)

~~Chapter 1: Databases and Database Users~~ (see the previous posting on Ch. 1 lectures)

Chapter 2: Database System Concepts and Architecture

Chapter 3: Data Modeling Using the Entity-Relationship (ER) Model.

The following questions on database systems concept and architecture discussed in Chapter 2 of Comp 3150 text book, Chapter 2, Comp 3150 posted course slide notes, are in-class questions for students to ponder and answer as I teach.

- The answers to the questions are found also by reviewing the Comp 3150, posted power point slide notes for Chapter 2 and being in class.
- Students are advised to review Chapters 1 to 3 of course book and Comp 3150 posted slide notes before and after each class.

- I will also go over the Slide notes in class with examples and integrate them into the class lectures, which are also posted in the More course material link on bright space.

1. WHAT ARE DATA MODELS, DATABASE SCHEMAS AND INSTANCES?

2. WHAT ARE THREE-SCHEMA ARCHITECTURE OF THE DATABASE?

3. WHAT DOES PROVISION OF DATA INDEPENDENCE MEAN?

4. WHAT ARE DBMS LANGUAGES AND INTERFACES?

5. WHAT ARE DATABASE SYSTEM UTILITIES AND TOOLS?

6. WHAT ARE CENTRALIZED AND CLIENT-SERVER DBMS ARCHITECTURES

7. WHAT IS THE HISTORY OF DATABASE DATA MODELS?

**

Try to answer the 7 questions above with an example with reference to a simple 3-table database of student-takes-course given in Chapter 1 lecture in class.

An Example simple database Problem.

You are asked to define a simple student record database for keeping track of student grades in courses they take.

Solution

1. An example data model is relational data model where data representing entities and their relationships are presented as relations (or tables)

An example database schema (showing the structure of each of its entities and relationships) for the Student-takes-course database is:

Student (Studid, Name, Class, Major)

Course (Cid, Ctitle, Credithr, Dept)

Takes (Studid, Cid, grade)

An instance of the database showing the contents of the database at a particular point in time is:

Student

<u>Studid</u>	Name	Class	Major
1	John Smith	1	CS
2	Patty Moore	2	Math

Course

<u>Cid</u>	Ctitle	Credithr	Dept
Comp1400	Intro to C	3	CS
Comp3150	Database	3	CS
Math1720	Calculus	3	Math

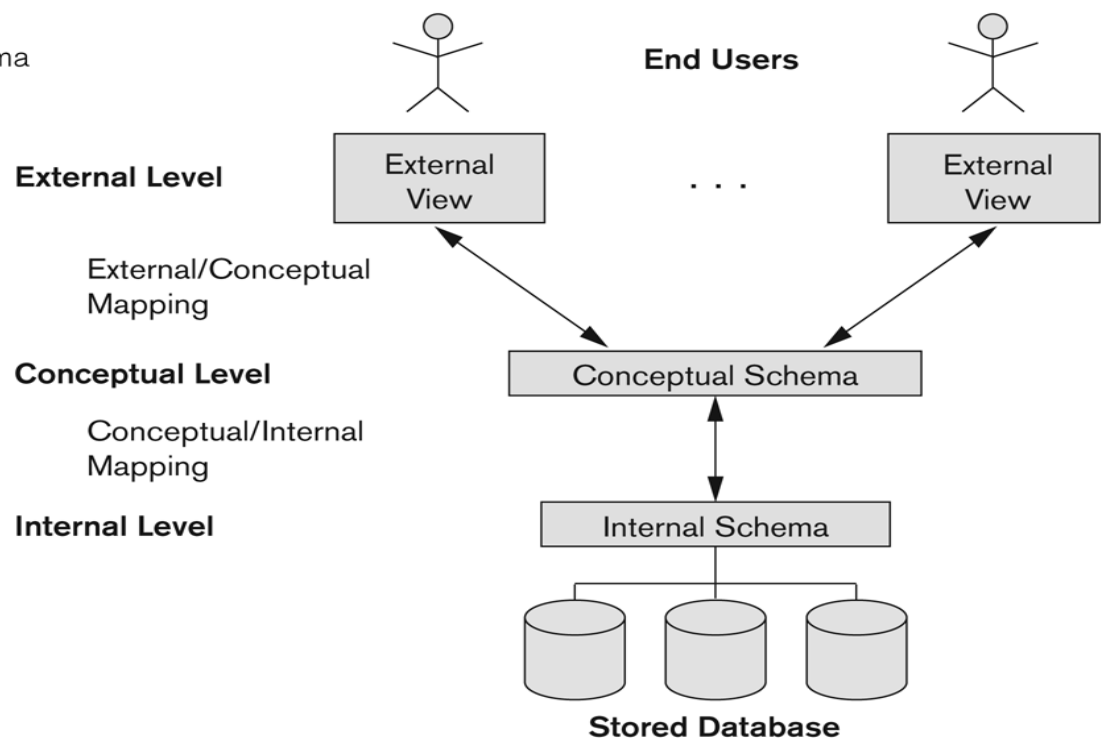
Takes

<u>Studid</u>	<u>Cid</u>	Grade
1	Comp1400	80
1	Math1720	70
2	Comp1400	60

2. Three-Schema Architecture

Figure 2.2

The three-schema architecture.



An example external level schema for user queries like “give me the name and studid of all CS students” is:
Student (Name, Studid).

Conceptual level Schema for Student in this example is:
Student (Studid: string (4), Name: string (20), Class:
Number (1), Major: string (4))

An example internal /physical storage format for a student record (that is the actual physical structure for

storing on disk the student table defined in the conceptual schema above (as given in Fig. 13, p13 of book) is:

Data item Name	record	Start position	length(bytes)
Studid	1		4
Name	5		20
Class	25		1
Major	26		4

- 3. What are data independence provided by the DBMS?

Slide 15 of Chapter 2 notes describes this as: This is the ability of the DBMS to change the schema at a lower level of DB system without having to change the schema at the next higher level.

- DBMS provides the following two types of data independence.

- 1. Logical data independence

The ability to change the conceptual schema (e.g. modify the schema of Student by adding address as an additional column) without having to change

the external schema or application program (which gets the name and studid)

- 2. Physical data independence: the ability to change the internal/physical schema without having to change the conceptual schema. For example, an access path (eg. Index) to improve retrieval speed of STUDENT records should not require a query like “list all students’ names and studid” to be changed.

1. DBMS LANGUAGES AND INTERFACES

Discussed on slides 18 to 22 of chapter 2 slide notes.

2. DATABASE SYSTEM UTILITIES AND TOOLS

Discussed on slides 23 to 25.

3. CENTRALIZED AND CLIENT-SERVER ARCHITECTURES.

Discussed on slides 26 and 27.

4. HISTORY OF DATA MODELS

Discussed on slide 28.

**

Chapter 3 Continues with Database Design through the Entity-Relationship model.

Slides 7 to 8 of Chapter 3 slides presents you with a requirements description of a COMPANY database and your job is to design the entity-relationship model database for this database.

- We need to create a database schema design based on the (simplified) **requirements** of the COMPANY Database presented on slides 7 and 8.
- Extract all the entities and their attributes. Also, extract all the relationship between entities and their attributes.
- Note that initially, some attributes of an entity may be an implicit relationship embedded with an entity description that can later be refined to be an explicit relationship in the final design. For example, in Department, Manager is an implicit relationship

which is later refined to a “Manages” explicit relationship.

- Identified entities are 4 with all initial attributes:
 1. DEPARTMENT (Name, Number, **Manager**, **Manager_start_date**, {Location})
 2. PROJECT (Name, Number, Location, Controlling_department)
 3. EMPLOYEE (SSN, Name (Fname, Minit, Lname), Address, Department, Birthdate, Sex, Supervisor, Salary, {Works_on(Project, Hours)})
 4. DEPENDENT (Dependent_name, Employee, Relationship, Birth_date, Sex)

- Relationships are 5 (now described in words but should also be clearly presented in Schema):
 1. An Employee manages Department
written in schema form as: **Manages(SSN, Number, Start_date)**
***** Note that this relationship Manages is an example of refining an implicit relationship attributes Manager, Manager_start_date in the Entity (1) schema Department above.**
 2. Each Department controls some PROJECTS.

This is written in schema form as:

Controls(DNumber, PNumber)

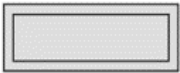
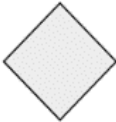
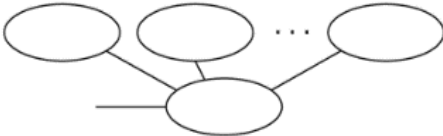
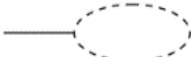
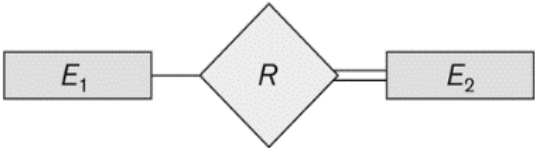
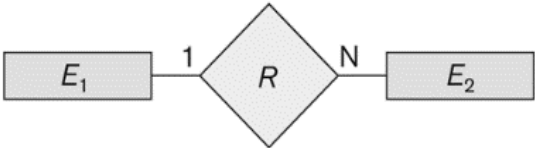
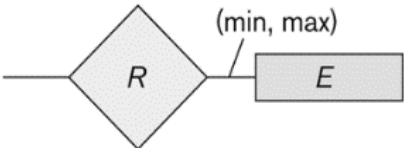
3. Each Employee works for one Department but may work on several PROJECTS. This is used in the explicit relationship **Works_For(Ssn, Dnumber)**.

4. Each Employee has a supervisor. This is a recursive relationship as: **Supervision(SSsn, ESsn)**.

5. Each Employee may have some DEPENDENTS. This is an identifying relationship for a weak entity type as: **Dependents_of(Ssn, Name)**

- Slide 19 of Chapter 3 presents a summary of symbols for representing different aspects of the model.

Figure 3.14
Summary of the
notation for ER
diagrams.

Symbol	Meaning
	Entity
	Weak Entity
	Relationship
	Identifying Relationship
	Attribute
	Key Attribute
	Multivalued Attribute
	Composite Attribute
	Derived Attribute
	Total Participation of E_2 in R
	Cardinality Ratio 1: N for $E_1:E_2$ in R
	Structural Constraint (min, max) on Participation of E in R

1. Each E2 record must relate to
An E1 record for total participation
2. Each E1 record can relate to up to
N records but each E2 record must relate to only 1 E1 record.
3. Each E1 record can relate to a minimum of 0 and maximum of N
E2 records. Each E2 record is related to 0 to 1 E1 records. Assume
we are interpreting the E1--1-- R -- N—E2 relation above using the
structural constraint of (min, max).

The full ER design for the COMPANY database is given

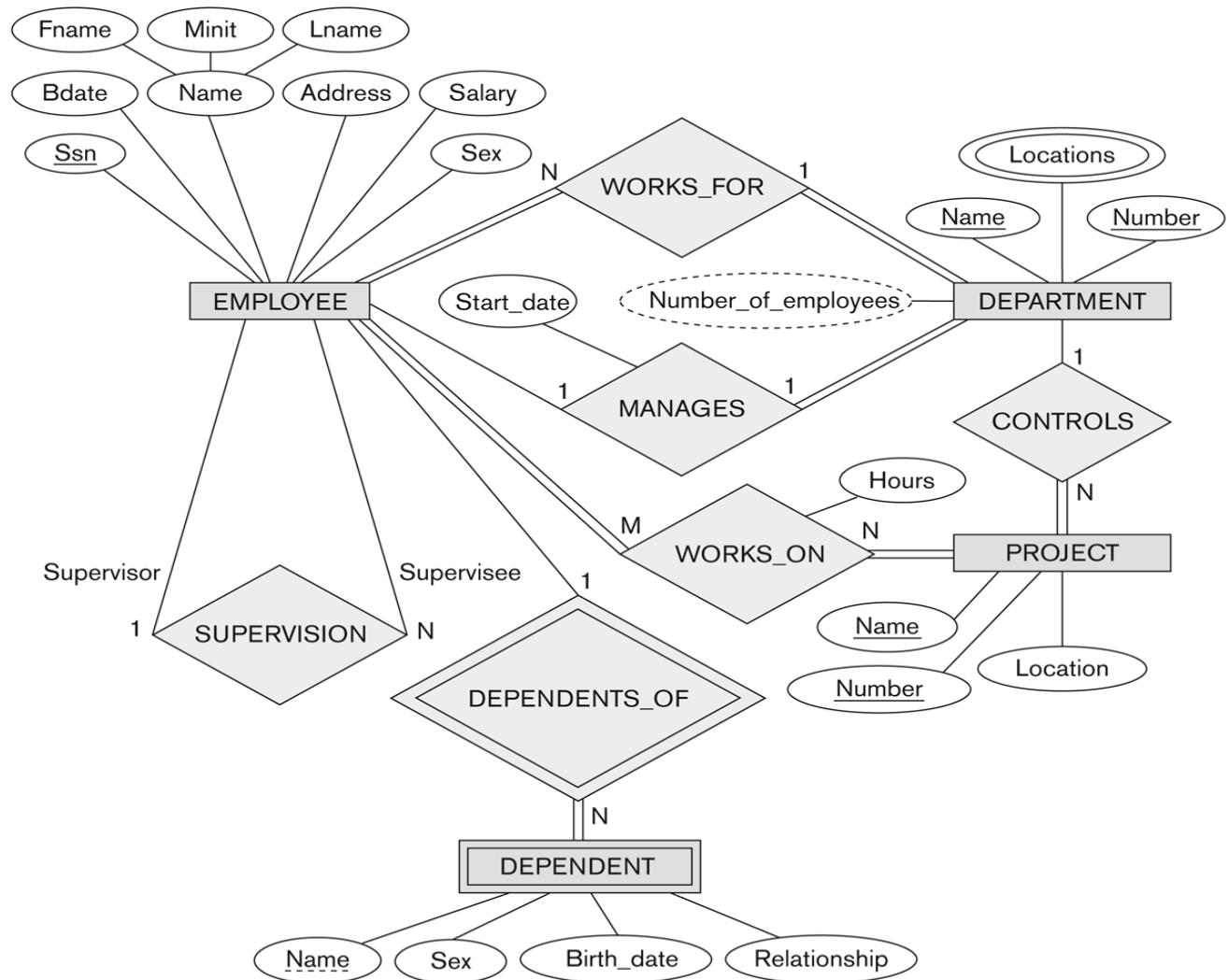
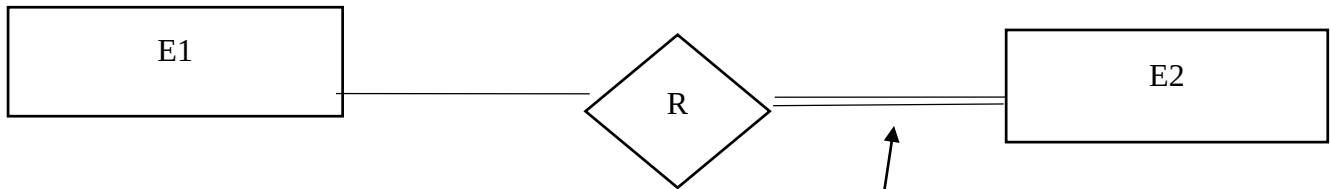


Figure 3.2

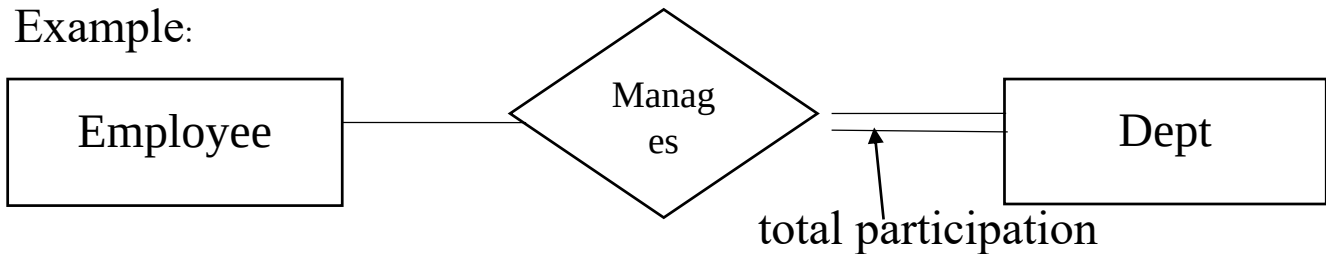
An ER schema diagram for the COMPANY database. The diagrammatic notation is introduced gradually throughout this chapter.

More on Discussion of Ch 3 ER Diagrams on slide 19

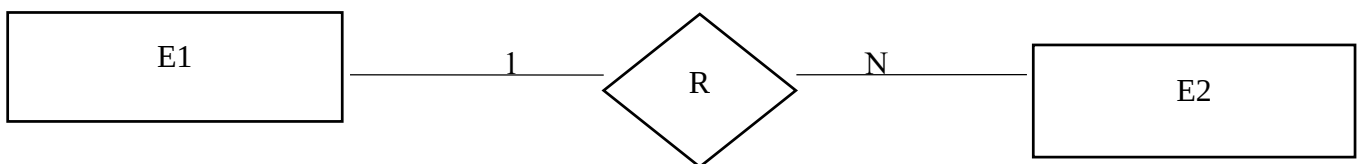


Means that every E2 record must relate to an E1 record through R
Shows total participation of entity E2 in relationship R.

Example:



1. For Total participation, Every dept is managed by an employee. (reading from right side entity to the left). (total)
2. However, not every employee is a manager of a dept. (reading from left to right side).



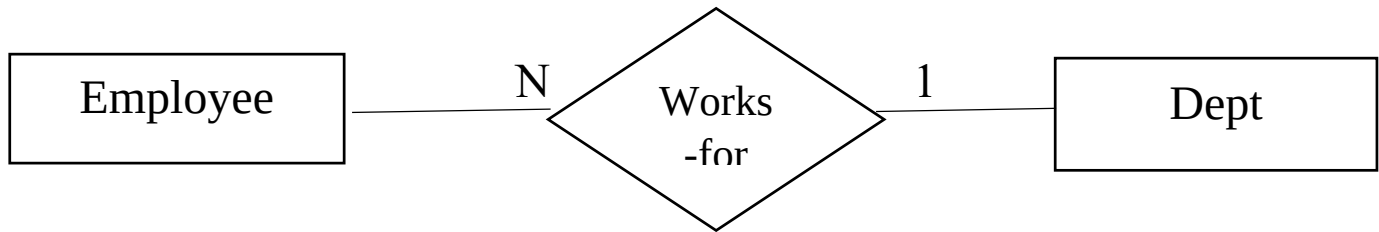
Shows Cardinality Ratio 1:N between entities E1:E2 through relationship R.

Meaning that:

1. Each E1 record is related to up to N records of E2. (reading from left side entity to right side entity).
2. Each E2 record is related to only 1 record of E1 through R. (reading from right side entity to left side).

ER Diagrams Slide 19 of Chapter 3 Notes.

Example. N:1 Relationship



Shows Cardinality Ratio

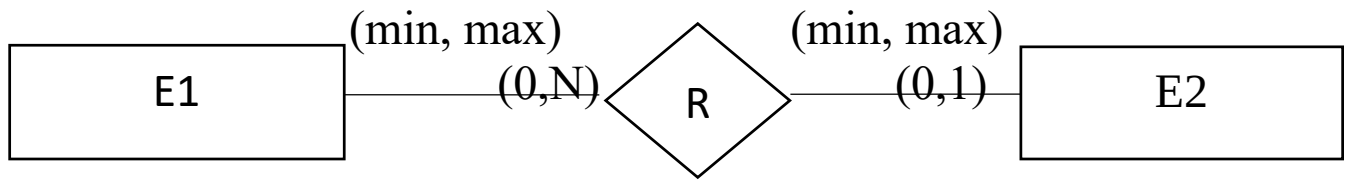
Meaning that :

1. Each employee record works for one dept. (reading from left side entity to right side entity).
2. However, each dept record has several employees (N) working for it. (reading from right side entity to left side).

An example database instance for this Employee-Works_for-Dept database relationship is (note there are 3 tables here):

<u>Employee</u>	<u>Works_for</u>		<u>Dept</u>
<u>Ssn</u>	<u>Ssn</u>	<u>Deptid</u>	<u>Deptid</u>
101	101	CS	CS
102	102	CS	Math
103	103	Math	Admin
104			

ER Diagrams on Slide 19. (Also, these alternative notations for structural constraints discussed on page 84 of course book).

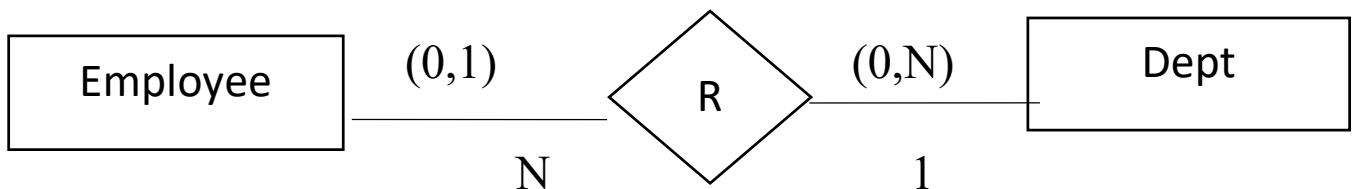


Shows structural constraints (min, max) on participation of E1 and E2 on R.

This means:

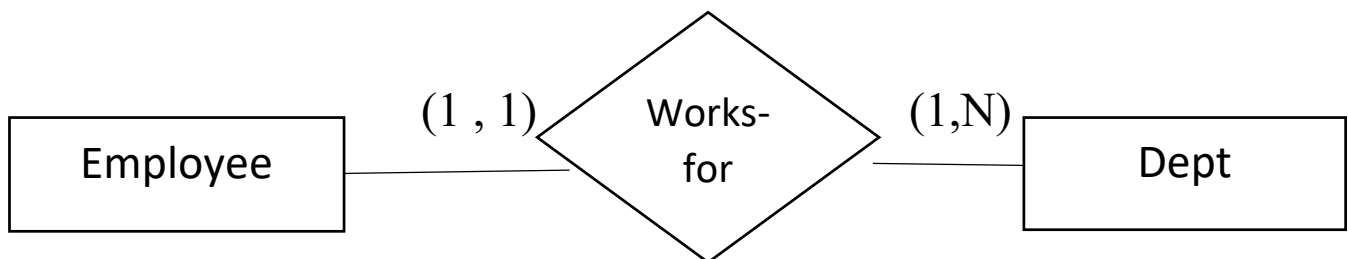
1. Each E1 record can relate to a minimum of 0 and a maximum of N of E2 records through R. (reading from left to right).
2. But each E2 record can relate to a minimum of 0 and a maximum of 1 records of E1 entity through R.

E.g.,



This means that each employee can work for 0 to 1 depts. However, each dept has 0 to N employees working for it.

To have total participation with both of these entities, we set the minimum to 1 as in:



Building the ER Diagram for the company DB.

From the requirements analysis, we define the entities with their attributes and relationships between the entities with their attributes.

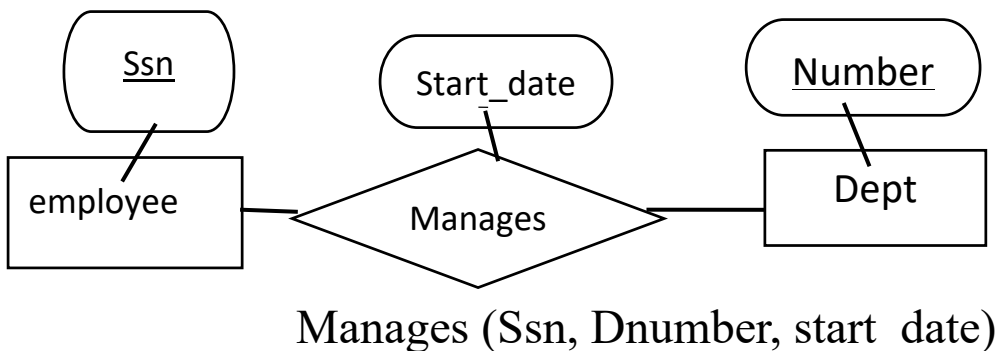
Entities:

1. DEPARTMENT (Name, Number, Locations, Manager, Manager_start_date)
Note: Location is a multivalued attribute.
2. PROJECT (Name, Number, Location, Controlling_dept)
3. EMPLOYEE (Name (Fname, Minit, Lname), Ssn, Sex, Address, Salary, Birth_date, Dept, Supervisor, Works_on)
4. DEPENDENT (Employee, Dependent_name, Sex, Birth_date,

Relationship)

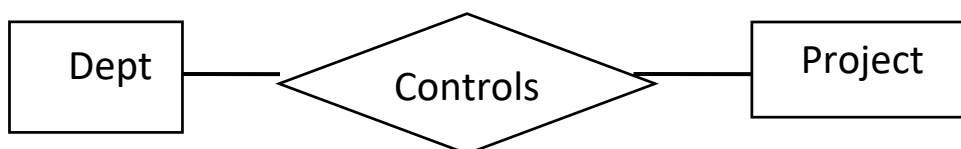
Relationships:

1. An Employee manages department.



2. Controls: Each Department controls some projects.

Controls (Dnumber, PNumber)



3. (a) Works_for
(b) Works_on

Each employee work for one Department but many work on several projects

Work_for(Ssn, Dnumber)

Work_on(Ssn, Pnumber, Hours)

4. Supervises:

Each employee has a supervisor.

Supervises(Ssn, eSsn)

5. Dependent_of:

Each employee may have some dependents.

Dependent_of (Ssn, Dependent_name)

Company Database Entity Relationship Model (Discussion on Refinement)

- Figure 3.8 on Ch. 3 slide notes, shows the preliminary design of the four entities types for company DB.
- In figure 3.8, some implicit relationship among entity types exist.
- This is when an attribute of one entity type refers to another entity type
- E.g. Attribute manager of Dept. refers to an employee who manages the Dept.
- In the ER model, these references should not be represented as attributes but relationships
- Thus, the initial company DB schema from fig. 3.8 is refined so that relationships are represented explicitly as relationships.
- The implicit relationships in Fig 3.8 refined into explicit relationships are:

1. Employee: Works_on Project (leads to removing the Works_on attribute previously in Employee during the initial design (see page 11 of this document, so it is now in a relationship type called Works_on).
 2. Department: is managed by Employee.
 3. Project : there is no implicit attribute to remove from Project
 4. Dependent: is of Employee
-
- After removing the implicit relationships from the entity type listed on the left above, we replace them with explicit relationship to connect the left entity type to the right hand entity type.
 - The final ER Diagram is Fig 3.2.
 - On ER, Cardinality ratios for binary relationship type are 1:1, 1:N, N:1 and M:N.
 - Two main types of binary relationship constraints are cardinality ration (e.g. 1:1) and participation (total or partial)
 - An alternative notation for specifying structural constraints shown in Fig. 3.15 (Slide 39) can be used to both the cardinality ratio and single/double line notation for participation constraint. (see slides 37, 38, 39, 19 for details).